

FRAME FOR PERFORMANCE

SSS 1-3 | Advanced Production Garden

Frame and execute a research or engineering mission on advanced production efficiency. Produce quantitative evidence that can support a design or management decision.

MISSION KIT production records, instruments, calculator

EVIDENCE yield component, harvest index or efficiency evidence

GM-401 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Frame with production records, instruments, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for yield component, harvest index or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Use documented methods. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Production Garden, including a measurable acceptance criterion.

ADVANCED RESEARCH MISSION

SSS 1-3 | Advanced Production Garden

Frame and execute a research or engineering mission on advanced production efficiency. Produce quantitative evidence that can support a design or management decision.

MISSION KIT production records, instruments, calculator

EVIDENCE yield component, harvest index or efficiency evidence

GM-402 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Design with production records, instruments, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for yield component, harvest index or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Use documented methods. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Production Garden, including a measurable acceptance criterion.

INSTRUMENT THE DESIGN

SSS 1-3 | Advanced Production Garden

Frame and execute a research or engineering mission on advanced production efficiency. Produce quantitative evidence that can support a design or management decision.

MISSION KIT production records, instruments, calculator

EVIDENCE yield component, harvest index or efficiency evidence

GM-403 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Instrument with production records, instruments, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for yield component, harvest index or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Use documented methods. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Production Garden, including a measurable acceptance criterion.

CALIBRATE FOR PERFORMANCE

SSS 1-3 | Advanced Production Garden

Frame and execute a research or engineering mission on advanced production efficiency. Produce quantitative evidence that can support a design or management decision.

MISSION KIT production records, instruments, calculator

EVIDENCE yield component, harvest index or efficiency evidence

GM-404 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Calibrate with production records, instruments, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for yield component, harvest index or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Use documented methods. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Production Garden, including a measurable acceptance criterion.

ADVANCED RESEARCH MISSION

SSS 1-3 | Advanced Production Garden

Frame and execute a research or engineering mission on advanced production efficiency. Produce quantitative evidence that can support a design or management decision.

MISSION KIT production records, instruments, calculator

EVIDENCE yield component, harvest index or efficiency evidence

GM-405 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Model with production records, instruments, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for yield component, harvest index or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Use documented methods. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Production Garden, including a measurable acceptance criterion.

VALIDATE THE DESIGN

SSS 1-3 | Advanced Production Garden

Frame and execute a research or engineering mission on advanced production efficiency. Produce quantitative evidence that can support a design or management decision.

MISSION KIT production records, instruments, calculator

EVIDENCE yield component, harvest index or efficiency evidence

GM-406 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Validate with production records, instruments, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for yield component, harvest index or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Use documented methods. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Production Garden, including a measurable acceptance criterion.

EVALUATE FOR PERFORMANCE

SSS 1-3 | Advanced Production Garden

Frame and execute a research or engineering mission on advanced production efficiency. Produce quantitative evidence that can support a design or management decision.

MISSION KIT production records, instruments, calculator

EVIDENCE yield component, harvest index or efficiency evidence

GM-407 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Evaluate with production records, instruments, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for yield component, harvest index or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Use documented methods. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Production Garden, including a measurable acceptance criterion.

ADVANCED RESEARCH MISSION

SSS 1-3 | Advanced Production Garden

Frame and execute a research or engineering mission on advanced production efficiency. Produce quantitative evidence that can support a design or management decision.

MISSION KIT production records, instruments, calculator

EVIDENCE yield component, harvest index or efficiency evidence

GM-408 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Optimize with production records, instruments, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for yield component, harvest index or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Use documented methods. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Production Garden, including a measurable acceptance criterion.

STRESS-TEST THE DESIGN

SSS 1-3 | Advanced Production Garden

Frame and execute a research or engineering mission on advanced production efficiency. Produce quantitative evidence that can support a design or management decision.

MISSION KIT production records, instruments, calculator

EVIDENCE yield component, harvest index or efficiency evidence

GM-409 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Stress-test with production records, instruments, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for yield component, harvest index or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Use documented methods. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Production Garden, including a measurable acceptance criterion.

RECOMMEND FOR PERFORMANCE

SSS 1-3 | Advanced Production Garden

Frame and execute a research or engineering mission on advanced production efficiency. Produce quantitative evidence that can support a design or management decision.

MISSION KIT production records, instruments, calculator

EVIDENCE yield component, harvest index or efficiency evidence

GM-410 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Recommend with production records, instruments, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for yield component, harvest index or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Use documented methods. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Production Garden, including a measurable acceptance criterion.

FRAME FOR PERFORMANCE

SSS 1-3 | Crop Trial Garden

Frame and execute a research or engineering mission on replicated crop trials. Produce quantitative evidence that can support a design or management decision.

MISSION KIT trial map, treatment labels, data sheet

EVIDENCE mean, SD, treatment effect or block evidence

GM-411 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Frame with trial map, treatment labels, data sheet. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for mean, SD, treatment effect or block evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Preserve treatment identity. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Crop Trial Garden, including a measurable acceptance criterion.

CROP RESEARCH MISSION

SSS 1-3 | Crop Trial Garden

Frame and execute a research or engineering mission on replicated crop trials. Produce quantitative evidence that can support a design or management decision.

MISSION KIT trial map, treatment labels, data sheet

EVIDENCE mean, SD, treatment effect or block evidence

GM-412 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Design with trial map, treatment labels, data sheet. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for mean, SD, treatment effect or block evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Preserve treatment identity. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Crop Trial Garden, including a measurable acceptance criterion.

INSTRUMENT THE DESIGN

SSS 1-3 | Crop Trial Garden

Frame and execute a research or engineering mission on replicated crop trials. Produce quantitative evidence that can support a design or management decision.

MISSION KIT trial map, treatment labels, data sheet

EVIDENCE mean, SD, treatment effect or block evidence

GM-413 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Instrument with trial map, treatment labels, data sheet. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for mean, SD, treatment effect or block evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Preserve treatment identity. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Crop Trial Garden, including a measurable acceptance criterion.

CALIBRATE FOR PERFORMANCE

SSS 1-3 | Crop Trial Garden

Frame and execute a research or engineering mission on replicated crop trials. Produce quantitative evidence that can support a design or management decision.

MISSION KIT trial map, treatment labels, data sheet

EVIDENCE mean, SD, treatment effect or block evidence

GM-414 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Calibrate with trial map, treatment labels, data sheet. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for mean, SD, treatment effect or block evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Preserve treatment identity. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Crop Trial Garden, including a measurable acceptance criterion.

CROP RESEARCH MISSION

SSS 1-3 | Crop Trial Garden

Frame and execute a research or engineering mission on replicated crop trials. Produce quantitative evidence that can support a design or management decision.

MISSION KIT trial map, treatment labels, data sheet

EVIDENCE mean, SD, treatment effect or block evidence

GM-415 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Model with trial map, treatment labels, data sheet. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for mean, SD, treatment effect or block evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Preserve treatment identity. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Crop Trial Garden, including a measurable acceptance criterion.

VALIDATE THE DESIGN

SSS 1-3 | Crop Trial Garden

Frame and execute a research or engineering mission on replicated crop trials. Produce quantitative evidence that can support a design or management decision.

MISSION KIT trial map, treatment labels, data sheet

EVIDENCE mean, SD, treatment effect or block evidence

GM-416 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Validate with trial map, treatment labels, data sheet. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for mean, SD, treatment effect or block evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Preserve treatment identity. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Crop Trial Garden, including a measurable acceptance criterion.

EVALUATE FOR PERFORMANCE

SSS 1-3 | Crop Trial Garden

Frame and execute a research or engineering mission on replicated crop trials. Produce quantitative evidence that can support a design or management decision.

MISSION KIT trial map, treatment labels, data sheet

EVIDENCE mean, SD, treatment effect or block evidence

GM-417 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Evaluate with trial map, treatment labels, data sheet. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for mean, SD, treatment effect or block evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Preserve treatment identity. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Crop Trial Garden, including a measurable acceptance criterion.

CROP RESEARCH MISSION

SSS 1-3 | Crop Trial Garden

Frame and execute a research or engineering mission on replicated crop trials. Produce quantitative evidence that can support a design or management decision.

MISSION KIT trial map, treatment labels, data sheet

EVIDENCE mean, SD, treatment effect or block evidence

GM-418 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Optimize with trial map, treatment labels, data sheet. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for mean, SD, treatment effect or block evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Preserve treatment identity. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Crop Trial Garden, including a measurable acceptance criterion.

STRESS-TEST THE DESIGN

SSS 1-3 | Crop Trial Garden

Frame and execute a research or engineering mission on replicated crop trials. Produce quantitative evidence that can support a design or management decision.

MISSION KIT trial map, treatment labels, data sheet

EVIDENCE mean, SD, treatment effect or block evidence

GM-419 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Stress-test with trial map, treatment labels, data sheet. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for mean, SD, treatment effect or block evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Preserve treatment identity. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Crop Trial Garden, including a measurable acceptance criterion.

RECOMMEND FOR PERFORMANCE

SSS 1-3 | Crop Trial Garden

Frame and execute a research or engineering mission on replicated crop trials. Produce quantitative evidence that can support a design or management decision.

MISSION KIT trial map, treatment labels, data sheet

EVIDENCE mean, SD, treatment effect or block evidence

GM-420 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Recommend with trial map, treatment labels, data sheet. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for mean, SD, treatment effect or block evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Preserve treatment identity. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Crop Trial Garden, including a measurable acceptance criterion.

FRAME FOR PERFORMANCE

SSS 1-3 | Controlled Experiment Garden

Frame and execute a research or engineering mission on causal experimental design. Produce quantitative evidence that can support a design or management decision.

MISSION KIT experimental layout, instruments

EVIDENCE factor effect, interaction, confounding or sensitivity evidence

GM-421 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Frame with experimental layout, instruments. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for factor effect, interaction, confounding or sensitivity evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Follow protocol exactly. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Controlled Experiment Garden, including a measurable acceptance criterion.

CONTROLLED RESEARCH MISSION

SSS 1-3 | Controlled Experiment Garden

Frame and execute a research or engineering mission on causal experimental design. Produce quantitative evidence that can support a design or management decision.

MISSION KIT experimental layout, instruments

EVIDENCE factor effect, interaction, confounding or sensitivity evidence

GM-422 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Design with experimental layout, instruments. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for factor effect, interaction, confounding or sensitivity evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Follow protocol exactly. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Controlled Experiment Garden, including a measurable acceptance criterion.

INSTRUMENT THE DESIGN

SSS 1-3 | Controlled Experiment Garden

Frame and execute a research or engineering mission on causal experimental design. Produce quantitative evidence that can support a design or management decision.

MISSION KIT experimental layout, instruments

EVIDENCE factor effect, interaction, confounding or sensitivity evidence

GM-423 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Instrument with experimental layout, instruments. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for factor effect, interaction, confounding or sensitivity evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Follow protocol exactly. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Controlled Experiment Garden, including a measurable acceptance criterion.

CALIBRATE FOR PERFORMANCE

SSS 1-3 | Controlled Experiment Garden

Frame and execute a research or engineering mission on causal experimental design. Produce quantitative evidence that can support a design or management decision.

MISSION KIT experimental layout, instruments

EVIDENCE factor effect, interaction, confounding or sensitivity evidence

GM-424 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Calibrate with experimental layout, instruments. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for factor effect, interaction, confounding or sensitivity evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Follow protocol exactly. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Controlled Experiment Garden, including a measurable acceptance criterion.

CONTROLLED RESEARCH MISSION

SSS 1-3 | Controlled Experiment Garden

Frame and execute a research or engineering mission on causal experimental design. Produce quantitative evidence that can support a design or management decision.

MISSION KIT experimental layout, instruments

EVIDENCE factor effect, interaction, confounding or sensitivity evidence

GM-425 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Model with experimental layout, instruments. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for factor effect, interaction, confounding or sensitivity evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Follow protocol exactly. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Controlled Experiment Garden, including a measurable acceptance criterion.

VALIDATE THE DESIGN

SSS 1-3 | Controlled Experiment Garden

Frame and execute a research or engineering mission on causal experimental design. Produce quantitative evidence that can support a design or management decision.

MISSION KIT experimental layout, instruments

EVIDENCE factor effect, interaction, confounding or sensitivity evidence

GM-426 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Validate with experimental layout, instruments. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for factor effect, interaction, confounding or sensitivity evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Follow protocol exactly. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Controlled Experiment Garden, including a measurable acceptance criterion.

EVALUATE FOR PERFORMANCE

SSS 1-3 | Controlled Experiment Garden

Frame and execute a research or engineering mission on causal experimental design. Produce quantitative evidence that can support a design or management decision.

MISSION KIT experimental layout, instruments

EVIDENCE factor effect, interaction, confounding or sensitivity evidence

GM-427 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Evaluate with experimental layout, instruments. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for factor effect, interaction, confounding or sensitivity evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Follow protocol exactly. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Controlled Experiment Garden, including a measurable acceptance criterion.

CONTROLLED RESEARCH MISSION

SSS 1-3 | Controlled Experiment Garden

Frame and execute a research or engineering mission on causal experimental design. Produce quantitative evidence that can support a design or management decision.

MISSION KIT experimental layout, instruments

EVIDENCE factor effect, interaction, confounding or sensitivity evidence

GM-428 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Optimize with experimental layout, instruments. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for factor effect, interaction, confounding or sensitivity evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Follow protocol exactly. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Controlled Experiment Garden, including a measurable acceptance criterion.

STRESS-TEST THE DESIGN

SSS 1-3 | Controlled Experiment Garden

Frame and execute a research or engineering mission on causal experimental design. Produce quantitative evidence that can support a design or management decision.

MISSION KIT experimental layout, instruments

EVIDENCE factor effect, interaction, confounding or sensitivity evidence

GM-429 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Stress-test with experimental layout, instruments. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for factor effect, interaction, confounding or sensitivity evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Follow protocol exactly. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Controlled Experiment Garden, including a measurable acceptance criterion.

RECOMMEND FOR PERFORMANCE

SSS 1-3 | Controlled Experiment Garden

Frame and execute a research or engineering mission on causal experimental design. Produce quantitative evidence that can support a design or management decision.

MISSION KIT experimental layout, instruments

EVIDENCE factor effect, interaction, confounding or sensitivity evidence

GM-430 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Recommend with experimental layout, instruments. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for factor effect, interaction, confounding or sensitivity evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Follow protocol exactly. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Controlled Experiment Garden, including a measurable acceptance criterion.

FRAME FOR PERFORMANCE

SSS 1-3 | Irrigation Engineering Garden

Frame and execute a research or engineering mission on hydraulic design and performance. Produce quantitative evidence that can support a design or management decision.

MISSION KIT flow tools, elevation/pipe data, calculator

EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-431 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Frame with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** No electrical or pressurized work without supervision. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Irrigation Engineering Garden, including a measurable acceptance criterion.

IRRIGATION RESEARCH MISSION

SSS 1-3 | Irrigation Engineering Garden

Frame and execute a research or engineering mission on hydraulic design and performance. Produce quantitative evidence that can support a design or management decision.

MISSION KIT flow tools, elevation/pipe data, calculator

EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-432 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Design with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** No electrical or pressurized work without supervision. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Irrigation Engineering Garden, including a measurable acceptance criterion.

INSTRUMENT THE DESIGN

SSS 1-3 | Irrigation Engineering Garden

Frame and execute a research or engineering mission on hydraulic design and performance. Produce quantitative evidence that can support a design or management decision.

MISSION KIT flow tools, elevation/pipe data, calculator

EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-433 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Instrument with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** No electrical or pressurized work without supervision. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Irrigation Engineering Garden, including a measurable acceptance criterion.

CALIBRATE FOR PERFORMANCE

SSS 1-3 | Irrigation Engineering Garden

Frame and execute a research or engineering mission on hydraulic design and performance. Produce quantitative evidence that can support a design or management decision.

MISSION KIT flow tools, elevation/pipe data, calculator

EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-434 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Calibrate with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** No electrical or pressurized work without supervision. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Irrigation Engineering Garden, including a measurable acceptance criterion.

IRRIGATION RESEARCH MISSION

SSS 1-3 | Irrigation Engineering Garden

Frame and execute a research or engineering mission on hydraulic design and performance. Produce quantitative evidence that can support a design or management decision.

MISSION KIT flow tools, elevation/pipe data, calculator

EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-435 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Model with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** No electrical or pressurized work without supervision. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Irrigation Engineering Garden, including a measurable acceptance criterion.

VALIDATE THE DESIGN

SSS 1-3 | Irrigation Engineering Garden

Frame and execute a research or engineering mission on hydraulic design and performance. Produce quantitative evidence that can support a design or management decision.

MISSION KIT flow tools, elevation/pipe data, calculator

EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-436 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Validate with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** No electrical or pressurized work without supervision. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Irrigation Engineering Garden, including a measurable acceptance criterion.

EVALUATE FOR PERFORMANCE

SSS 1-3 | Irrigation Engineering Garden

Frame and execute a research or engineering mission on hydraulic design and performance. Produce quantitative evidence that can support a design or management decision.

MISSION KIT flow tools, elevation/pipe data, calculator

EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-437 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Evaluate with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** No electrical or pressurized work without supervision. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Irrigation Engineering Garden, including a measurable acceptance criterion.

IRRIGATION RESEARCH MISSION

SSS 1-3 | Irrigation Engineering Garden

Frame and execute a research or engineering mission on hydraulic design and performance. Produce quantitative evidence that can support a design or management decision.

MISSION KIT flow tools, elevation/pipe data, calculator

EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-438 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Optimize with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** No electrical or pressurized work without supervision. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Irrigation Engineering Garden, including a measurable acceptance criterion.

STRESS-TEST THE DESIGN

SSS 1-3 | Irrigation Engineering Garden

Frame and execute a research or engineering mission on hydraulic design and performance. Produce quantitative evidence that can support a design or management decision.

MISSION KIT flow tools, elevation/pipe data, calculator

EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-439 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Stress-test with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** No electrical or pressurized work without supervision. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Irrigation Engineering Garden, including a measurable acceptance criterion.

RECOMMEND FOR PERFORMANCE

SSS 1-3 | Irrigation Engineering Garden

Frame and execute a research or engineering mission on hydraulic design and performance. Produce quantitative evidence that can support a design or management decision.

MISSION KIT flow tools, elevation/pipe data, calculator

EVIDENCE flow, head, loss, uniformity or efficiency evidence

GM-440 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Recommend with flow tools, elevation/pipe data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for flow, head, loss, uniformity or efficiency evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** No electrical or pressurized work without supervision. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Irrigation Engineering Garden, including a measurable acceptance criterion.

FRAME FOR PERFORMANCE

SSS 1-3 | Water Productivity Garden

Frame and execute a research or engineering mission on water-balance and productivity analysis. Produce quantitative evidence that can support a design or management decision.

MISSION KIT water records, yield data, calculator

EVIDENCE water balance, ET, IWP or ETWP evidence

GM-441 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Frame with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** State data sources and assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Water Productivity Garden, including a measurable acceptance criterion.

WATER RESEARCH MISSION

SSS 1-3 | Water Productivity Garden

Frame and execute a research or engineering mission on water-balance and productivity analysis. Produce quantitative evidence that can support a design or management decision.

MISSION KIT water records, yield data, calculator

EVIDENCE water balance, ET, IWP or ETWP evidence

GM-442 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Design with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** State data sources and assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Water Productivity Garden, including a measurable acceptance criterion.

INSTRUMENT THE DESIGN

SSS 1-3 | Water Productivity Garden

Frame and execute a research or engineering mission on water-balance and productivity analysis. Produce quantitative evidence that can support a design or management decision.

MISSION KIT water records, yield data, calculator

EVIDENCE water balance, ET, IWP or ETWP evidence

GM-443 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Instrument with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** State data sources and assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Water Productivity Garden, including a measurable acceptance criterion.

CALIBRATE FOR PERFORMANCE

SSS 1-3 | Water Productivity Garden

Frame and execute a research or engineering mission on water-balance and productivity analysis. Produce quantitative evidence that can support a design or management decision.

MISSION KIT water records, yield data, calculator

EVIDENCE water balance, ET, IWP or ETWP evidence

GM-444 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Calibrate with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** State data sources and assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Water Productivity Garden, including a measurable acceptance criterion.

WATER RESEARCH MISSION

SSS 1-3 | Water Productivity Garden

Frame and execute a research or engineering mission on water-balance and productivity analysis. Produce quantitative evidence that can support a design or management decision.

MISSION KIT water records, yield data, calculator

EVIDENCE water balance, ET, IWP or ETWP evidence

GM-445 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Model with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** State data sources and assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Water Productivity Garden, including a measurable acceptance criterion.

VALIDATE THE DESIGN

SSS 1-3 | Water Productivity Garden

Frame and execute a research or engineering mission on water-balance and productivity analysis. Produce quantitative evidence that can support a design or management decision.

MISSION KIT water records, yield data, calculator

EVIDENCE water balance, ET, IWP or ETWP evidence

GM-446 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Validate with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** State data sources and assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Water Productivity Garden, including a measurable acceptance criterion.

EVALUATE FOR PERFORMANCE

SSS 1-3 | Water Productivity Garden

Frame and execute a research or engineering mission on water-balance and productivity analysis. Produce quantitative evidence that can support a design or management decision.

MISSION KIT water records, yield data, calculator

EVIDENCE water balance, ET, IWP or ETWP evidence

GM-447 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Evaluate with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** State data sources and assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Water Productivity Garden, including a measurable acceptance criterion.

WATER RESEARCH MISSION

SSS 1-3 | Water Productivity Garden

Frame and execute a research or engineering mission on water-balance and productivity analysis. Produce quantitative evidence that can support a design or management decision.

MISSION KIT water records, yield data, calculator

EVIDENCE water balance, ET, IWP or ETWP evidence

GM-448 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Optimize with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** State data sources and assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Water Productivity Garden, including a measurable acceptance criterion.

STRESS-TEST THE DESIGN

SSS 1-3 | Water Productivity Garden

Frame and execute a research or engineering mission on water-balance and productivity analysis. Produce quantitative evidence that can support a design or management decision.

MISSION KIT water records, yield data, calculator

EVIDENCE water balance, ET, IWP or ETWP evidence

GM-449 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Stress-test with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** State data sources and assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Water Productivity Garden, including a measurable acceptance criterion.

RECOMMEND FOR PERFORMANCE

SSS 1-3 | Water Productivity Garden

Frame and execute a research or engineering mission on water-balance and productivity analysis. Produce quantitative evidence that can support a design or management decision.

MISSION KIT water records, yield data, calculator

EVIDENCE water balance, ET, IWP or ETWP evidence

GM-450 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Recommend with water records, yield data, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for water balance, ET, IWP or ETWP evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** State data sources and assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Water Productivity Garden, including a measurable acceptance criterion.

FRAME FOR PERFORMANCE

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-451 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Frame with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

ADVANCED RESEARCH MISSION

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-452 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Design with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

INSTRUMENT THE DESIGN

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-453 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Instrument with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

CALIBRATE FOR PERFORMANCE

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-454 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Calibrate with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

ADVANCED RESEARCH MISSION

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-455 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Model with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

VALIDATE THE DESIGN

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-456 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Validate with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

EVALUATE FOR PERFORMANCE

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-457 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Evaluate with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

ADVANCED RESEARCH MISSION

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-458 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Optimize with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

STRESS-TEST THE DESIGN

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-459 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Stress-test with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

RECOMMEND FOR PERFORMANCE

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-460 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Recommend with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

FRAME FOR PERFORMANCE

SSS 1-3 | Sensor & Monitoring Garden

Frame and execute a research or engineering mission on instrumentation and monitoring quality. Produce quantitative evidence that can support a design or management decision.

MISSION KIT sensor, reference, calibration data

EVIDENCE bias, precision, calibration, MAE/RMSE or placement evidence

GM-461 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Frame with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Protect instruments. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Sensor & Monitoring Garden, including a measurable acceptance criterion.

SENSOR RESEARCH MISSION

SSS 1-3 | Sensor & Monitoring Garden

Frame and execute a research or engineering mission on instrumentation and monitoring quality. Produce quantitative evidence that can support a design or management decision.

MISSION KIT sensor, reference, calibration data

EVIDENCE bias, precision, calibration, MAE/RMSE or placement evidence

GM-462 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Design with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Protect instruments. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Sensor & Monitoring Garden, including a measurable acceptance criterion.

INSTRUMENT THE DESIGN

SSS 1-3 | Sensor & Monitoring Garden

Frame and execute a research or engineering mission on instrumentation and monitoring quality. Produce quantitative evidence that can support a design or management decision.

MISSION KIT sensor, reference, calibration data

EVIDENCE bias, precision, calibration, MAE/RMSE or placement evidence

GM-463 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Instrument with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Protect instruments. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Sensor & Monitoring Garden, including a measurable acceptance criterion.

CALIBRATE FOR PERFORMANCE

SSS 1-3 | Sensor & Monitoring Garden

Frame and execute a research or engineering mission on instrumentation and monitoring quality. Produce quantitative evidence that can support a design or management decision.

MISSION KIT sensor, reference, calibration data

EVIDENCE bias, precision, calibration, MAE/RMSE or placement evidence

GM-464 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Calibrate with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Protect instruments. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Sensor & Monitoring Garden, including a measurable acceptance criterion.

SENSOR RESEARCH MISSION

SSS 1-3 | Sensor & Monitoring Garden

Frame and execute a research or engineering mission on instrumentation and monitoring quality. Produce quantitative evidence that can support a design or management decision.

MISSION KIT sensor, reference, calibration data

EVIDENCE bias, precision, calibration, MAE/RMSE or placement evidence

GM-465 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Model with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Protect instruments. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Sensor & Monitoring Garden, including a measurable acceptance criterion.

VALIDATE THE DESIGN

SSS 1-3 | Sensor & Monitoring Garden

Frame and execute a research or engineering mission on instrumentation and monitoring quality. Produce quantitative evidence that can support a design or management decision.

MISSION KIT sensor, reference, calibration data

EVIDENCE bias, precision, calibration, MAE/RMSE or placement evidence

GM-466 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Validate with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Protect instruments. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Sensor & Monitoring Garden, including a measurable acceptance criterion.

EVALUATE FOR PERFORMANCE

SSS 1-3 | Sensor & Monitoring Garden

Frame and execute a research or engineering mission on instrumentation and monitoring quality. Produce quantitative evidence that can support a design or management decision.

MISSION KIT sensor, reference, calibration data

EVIDENCE bias, precision, calibration, MAE/RMSE or placement evidence

GM-467 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Evaluate with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Protect instruments. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Sensor & Monitoring Garden, including a measurable acceptance criterion.

SENSOR RESEARCH MISSION

SSS 1-3 | Sensor & Monitoring Garden

Frame and execute a research or engineering mission on instrumentation and monitoring quality. Produce quantitative evidence that can support a design or management decision.

MISSION KIT sensor, reference, calibration data

EVIDENCE bias, precision, calibration, MAE/RMSE or placement evidence

GM-468 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Optimize with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Protect instruments. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Sensor & Monitoring Garden, including a measurable acceptance criterion.

STRESS-TEST THE DESIGN

SSS 1-3 | Sensor & Monitoring Garden

Frame and execute a research or engineering mission on instrumentation and monitoring quality. Produce quantitative evidence that can support a design or management decision.

MISSION KIT sensor, reference, calibration data

EVIDENCE bias, precision, calibration, MAE/RMSE or placement evidence

GM-469 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Stress-test with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Protect instruments. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Sensor & Monitoring Garden, including a measurable acceptance criterion.

RECOMMEND FOR PERFORMANCE

SSS 1-3 | Sensor & Monitoring Garden

Frame and execute a research or engineering mission on instrumentation and monitoring quality. Produce quantitative evidence that can support a design or management decision.

MISSION KIT sensor, reference, calibration data

EVIDENCE bias, precision, calibration, MAE/RMSE or placement evidence

GM-470 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Recommend with sensor, reference, calibration data. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for bias, precision, calibration, MAE/RMSE or placement evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Protect instruments. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Sensor & Monitoring Garden, including a measurable acceptance criterion.

FRAME FOR PERFORMANCE

SSS 1-3 | Smart Irrigation Demonstrator

Frame and execute a research or engineering mission on feedback irrigation control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-471 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Frame with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

SMART RESEARCH MISSION

SSS 1-3 | Smart Irrigation Demonstrator

Frame and execute a research or engineering mission on feedback irrigation control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-472 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Design with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

INSTRUMENT THE DESIGN

SSS 1-3 | Smart Irrigation Demonstrator

Frame and execute a research or engineering mission on feedback irrigation control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-473 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Instrument with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

CALIBRATE FOR PERFORMANCE

SSS 1-3 | Smart Irrigation Demonstrator

Frame and execute a research or engineering mission on feedback irrigation control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-474 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Calibrate with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

SMART RESEARCH MISSION

SSS 1-3 | Smart Irrigation Demonstrator

Frame and execute a research or engineering mission on feedback irrigation control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-475 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Model with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

VALIDATE THE DESIGN

SSS 1-3 | Smart Irrigation Demonstrator

Frame and execute a research or engineering mission on feedback irrigation control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-476 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Validate with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

EVALUATE FOR PERFORMANCE

SSS 1-3 | Smart Irrigation Demonstrator

Frame and execute a research or engineering mission on feedback irrigation control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-477 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Evaluate with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

SMART RESEARCH MISSION

SSS 1-3 | Smart Irrigation Demonstrator

Frame and execute a research or engineering mission on feedback irrigation control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-478 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Optimize with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

STRESS-TEST THE DESIGN

SSS 1-3 | Smart Irrigation Demonstrator

Frame and execute a research or engineering mission on feedback irrigation control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-479 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Stress-test with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

RECOMMEND FOR PERFORMANCE

SSS 1-3 | Smart Irrigation Demonstrator

Frame and execute a research or engineering mission on feedback irrigation control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-480 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Recommend with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

FRAME FOR PERFORMANCE

SSS 1-3 | Research & Design Garden

Frame and execute a research or engineering mission on engineering design decisions. Produce quantitative evidence that can support a design or management decision.

MISSION KIT requirements sheet, decision matrix

EVIDENCE requirement, score, test or techno-economic evidence

GM-481 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Frame with requirements sheet, decision matrix. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for requirement, score, test or techno-economic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Document assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Research & Design Garden, including a measurable acceptance criterion.

RESEARCH RESEARCH MISSION

SSS 1-3 | Research & Design Garden

Frame and execute a research or engineering mission on engineering design decisions. Produce quantitative evidence that can support a design or management decision.

MISSION KIT requirements sheet, decision matrix

EVIDENCE requirement, score, test or techno-economic evidence

GM-482 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Design with requirements sheet, decision matrix. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for requirement, score, test or techno-economic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Document assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Research & Design Garden, including a measurable acceptance criterion.

INSTRUMENT THE DESIGN

SSS 1-3 | Research & Design Garden

Frame and execute a research or engineering mission on engineering design decisions. Produce quantitative evidence that can support a design or management decision.

MISSION KIT requirements sheet, decision matrix

EVIDENCE requirement, score, test or techno-economic evidence

GM-483 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Instrument with requirements sheet, decision matrix. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for requirement, score, test or techno-economic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Document assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Research & Design Garden, including a measurable acceptance criterion.

CALIBRATE FOR PERFORMANCE

SSS 1-3 | Research & Design Garden

Frame and execute a research or engineering mission on engineering design decisions. Produce quantitative evidence that can support a design or management decision.

MISSION KIT requirements sheet, decision matrix

EVIDENCE requirement, score, test or techno-economic evidence

GM-484 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Calibrate with requirements sheet, decision matrix. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for requirement, score, test or techno-economic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Document assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Research & Design Garden, including a measurable acceptance criterion.

RESEARCH RESEARCH MISSION

SSS 1-3 | Research & Design Garden

Frame and execute a research or engineering mission on engineering design decisions. Produce quantitative evidence that can support a design or management decision.

MISSION KIT requirements sheet, decision matrix

EVIDENCE requirement, score, test or techno-economic evidence

GM-485 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Model with requirements sheet, decision matrix. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for requirement, score, test or techno-economic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Document assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Research & Design Garden, including a measurable acceptance criterion.

VALIDATE THE DESIGN

SSS 1-3 | Research & Design Garden

Frame and execute a research or engineering mission on engineering design decisions. Produce quantitative evidence that can support a design or management decision.

MISSION KIT requirements sheet, decision matrix

EVIDENCE requirement, score, test or techno-economic evidence

GM-486 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Validate with requirements sheet, decision matrix. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for requirement, score, test or techno-economic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Document assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Research & Design Garden, including a measurable acceptance criterion.

EVALUATE FOR PERFORMANCE

SSS 1-3 | Research & Design Garden

Frame and execute a research or engineering mission on engineering design decisions. Produce quantitative evidence that can support a design or management decision.

MISSION KIT requirements sheet, decision matrix

EVIDENCE requirement, score, test or techno-economic evidence

GM-487 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Evaluate with requirements sheet, decision matrix. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for requirement, score, test or techno-economic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Document assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Research & Design Garden, including a measurable acceptance criterion.

RESEARCH RESEARCH MISSION

SSS 1-3 | Research & Design Garden

Frame and execute a research or engineering mission on engineering design decisions. Produce quantitative evidence that can support a design or management decision.

MISSION KIT requirements sheet, decision matrix

EVIDENCE requirement, score, test or techno-economic evidence

GM-488 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Optimize with requirements sheet, decision matrix. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for requirement, score, test or techno-economic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Document assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Research & Design Garden, including a measurable acceptance criterion.

STRESS-TEST THE DESIGN

SSS 1-3 | Research & Design Garden

Frame and execute a research or engineering mission on engineering design decisions. Produce quantitative evidence that can support a design or management decision.

MISSION KIT requirements sheet, decision matrix

EVIDENCE requirement, score, test or techno-economic evidence

GM-489 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Stress-test with requirements sheet, decision matrix. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for requirement, score, test or techno-economic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Document assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Research & Design Garden, including a measurable acceptance criterion.

RECOMMEND FOR PERFORMANCE

SSS 1-3 | Research & Design Garden

Frame and execute a research or engineering mission on engineering design decisions. Produce quantitative evidence that can support a design or management decision.

MISSION KIT requirements sheet, decision matrix

EVIDENCE requirement, score, test or techno-economic evidence

GM-490 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Recommend with requirements sheet, decision matrix. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for requirement, score, test or techno-economic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Document assumptions. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Research & Design Garden, including a measurable acceptance criterion.

FRAME FOR PERFORMANCE

SSS 1-3 | Integrated Food Systems Laboratory

Frame and execute a research or engineering mission on integrated food-water-energy system behavior. Produce quantitative evidence that can support a design or management decision.

MISSION KIT system map, time series, calculator

EVIDENCE balance, mismatch, cascade or scenario evidence

GM-491 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Frame with system map, time series, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for balance, mismatch, cascade or scenario evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Respect laboratory procedures. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Integrated Food Systems Laboratory, including a measurable acceptance criterion.

INTEGRATED RESEARCH MISSION

SSS 1-3 | Integrated Food Systems Laboratory

Frame and execute a research or engineering mission on integrated food-water-energy system behavior. Produce quantitative evidence that can support a design or management decision.

MISSION KIT system map, time series, calculator

EVIDENCE balance, mismatch, cascade or scenario evidence

GM-492 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Design with system map, time series, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for balance, mismatch, cascade or scenario evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Respect laboratory procedures. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Integrated Food Systems Laboratory, including a measurable acceptance criterion.

INSTRUMENT THE DESIGN

SSS 1-3 | Integrated Food Systems Laboratory

Frame and execute a research or engineering mission on integrated food-water-energy system behavior. Produce quantitative evidence that can support a design or management decision.

MISSION KIT system map, time series, calculator

EVIDENCE balance, mismatch, cascade or scenario evidence

GM-493 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Instrument with system map, time series, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for balance, mismatch, cascade or scenario evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Respect laboratory procedures. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Integrated Food Systems Laboratory, including a measurable acceptance criterion.

CALIBRATE FOR PERFORMANCE

SSS 1-3 | Integrated Food Systems Laboratory

Frame and execute a research or engineering mission on integrated food-water-energy system behavior. Produce quantitative evidence that can support a design or management decision.

MISSION KIT system map, time series, calculator

EVIDENCE balance, mismatch, cascade or scenario evidence

GM-494 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Calibrate with system map, time series, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for balance, mismatch, cascade or scenario evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Respect laboratory procedures. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Integrated Food Systems Laboratory, including a measurable acceptance criterion.

INTEGRATED RESEARCH MISSION

SSS 1-3 | Integrated Food Systems Laboratory

Frame and execute a research or engineering mission on integrated food-water-energy system behavior. Produce quantitative evidence that can support a design or management decision.

MISSION KIT system map, time series, calculator

EVIDENCE balance, mismatch, cascade or scenario evidence

GM-495 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Model with system map, time series, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for balance, mismatch, cascade or scenario evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Respect laboratory procedures. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Integrated Food Systems Laboratory, including a measurable acceptance criterion.

VALIDATE THE DESIGN

SSS 1-3 | Integrated Food Systems Laboratory

Frame and execute a research or engineering mission on integrated food-water-energy system behavior. Produce quantitative evidence that can support a design or management decision.

MISSION KIT system map, time series, calculator

EVIDENCE balance, mismatch, cascade or scenario evidence

GM-496 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Validate with system map, time series, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for balance, mismatch, cascade or scenario evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Respect laboratory procedures. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Integrated Food Systems Laboratory, including a measurable acceptance criterion.

EVALUATE FOR PERFORMANCE

SSS 1-3 | Integrated Food Systems Laboratory

Frame and execute a research or engineering mission on integrated food-water-energy system behavior. Produce quantitative evidence that can support a design or management decision.

MISSION KIT system map, time series, calculator

EVIDENCE balance, mismatch, cascade or scenario evidence

GM-497 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Evaluate with system map, time series, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for balance, mismatch, cascade or scenario evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Respect laboratory procedures. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Integrated Food Systems Laboratory, including a measurable acceptance criterion.

INTEGRATED RESEARCH MISSION

SSS 1-3 | Integrated Food Systems Laboratory

Frame and execute a research or engineering mission on integrated food-water-energy system behavior. Produce quantitative evidence that can support a design or management decision.

MISSION KIT system map, time series, calculator

EVIDENCE balance, mismatch, cascade or scenario evidence

GM-498 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Optimize with system map, time series, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for balance, mismatch, cascade or scenario evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Respect laboratory procedures. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Integrated Food Systems Laboratory, including a measurable acceptance criterion.

STRESS-TEST THE DESIGN

SSS 1-3 | Integrated Food Systems Laboratory

Frame and execute a research or engineering mission on integrated food-water-energy system behavior. Produce quantitative evidence that can support a design or management decision.

MISSION KIT system map, time series, calculator

EVIDENCE balance, mismatch, cascade or scenario evidence

GM-499 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Stress-test with system map, time series, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for balance, mismatch, cascade or scenario evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Respect laboratory procedures. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Integrated Food Systems Laboratory, including a measurable acceptance criterion.

RECOMMEND FOR PERFORMANCE

SSS 1-3 | Integrated Food Systems Laboratory

Frame and execute a research or engineering mission on integrated food-water-energy system behavior. Produce quantitative evidence that can support a design or management decision.

MISSION KIT system map, time series, calculator

EVIDENCE balance, mismatch, cascade or scenario evidence

GM-500 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Recommend with system map, time series, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for balance, mismatch, cascade or scenario evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Respect laboratory procedures. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Integrated Food Systems Laboratory, including a measurable acceptance criterion.